



Smart Hire

TRUSTED LOCAL SERVICES, ONE CLICK AWAY

A service marketplace, built end to end.

Customers find and book verified local providers; providers manage jobs, availability and earnings — with an AI assistant helping both sides.

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PROJECT OVERVIEW

Finding a reliable local service shouldn't be a gamble.

Smart Hire connects customers with verified providers across categories like cleaning, tutoring, repairs and electrical work — with real bookings, real payments, real reviews, and an AI layer on top.



DATABASE

16

Tables, fully normalised



BACKEND

77

Live API endpoints



TESTING

48

Automated tests (front + back)



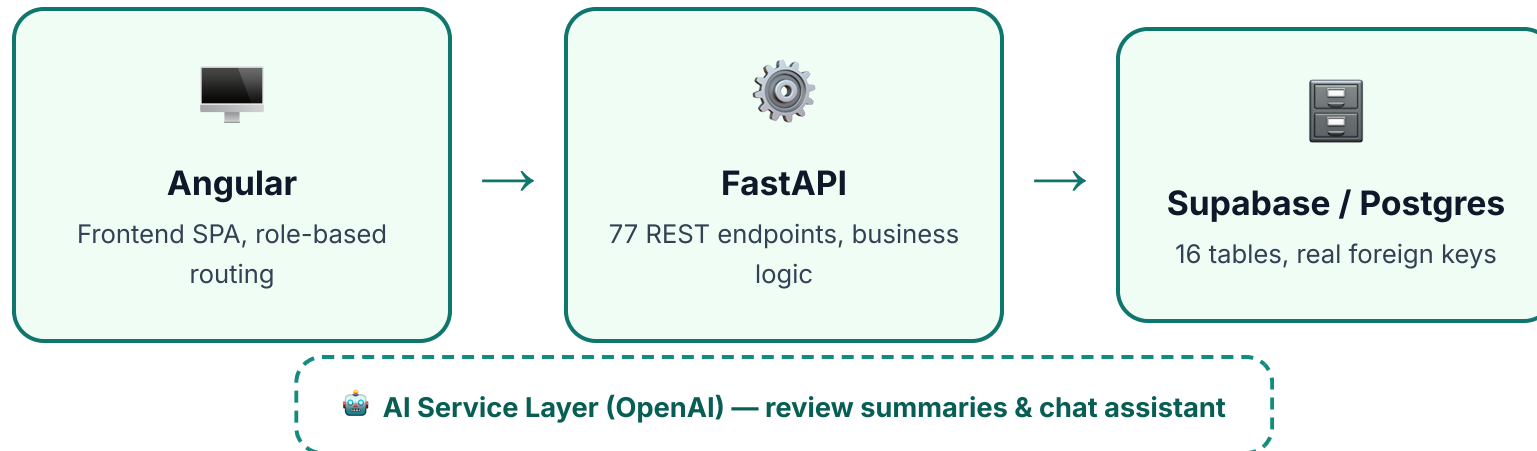
QUALITY

8

Real bugs found & fixed

Three layers, one clean data flow.

An Angular single-page app talks to a FastAPI backend over REST, which reads and writes a PostgreSQL database hosted on Supabase — with an AI service layer for review summaries and assistant chat.



One login, two very different workspaces.

The same account system serves customers and providers with completely different dashboards, guarded by role-based routing.



Role-based signup & login

Customer and provider registration, with route guards that keep each role in its own area.



Account-status handling

Suspended or pending accounts are redirected and blocked, not silently let through.



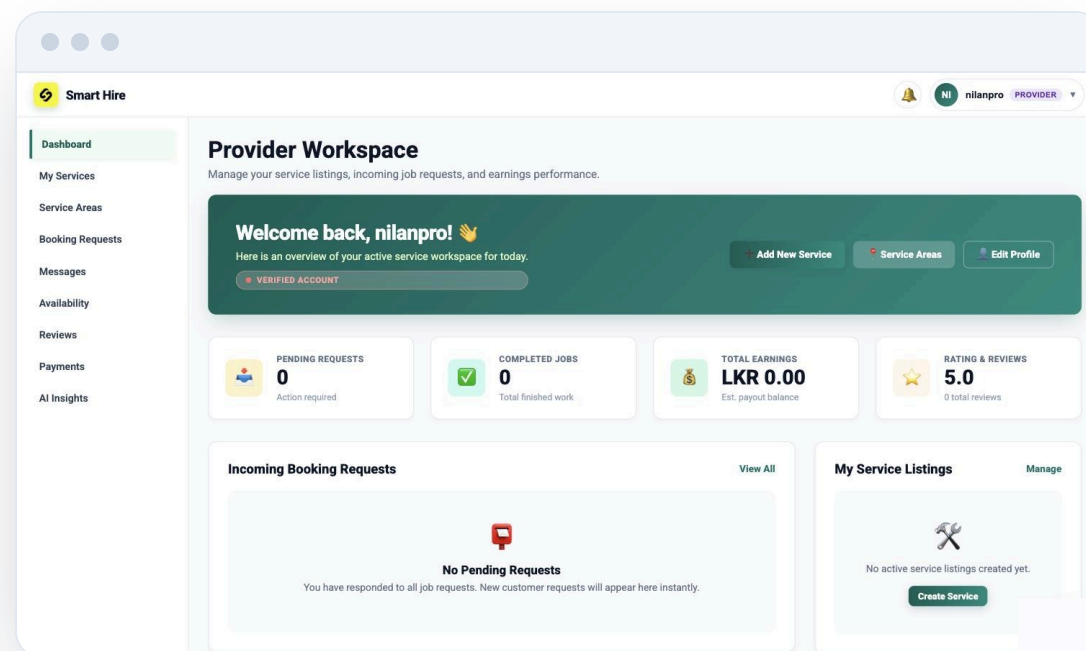
Editable profiles

Customer profiles and provider profiles (bio, experience, verification status) as first-class data.



Purpose-built dashboards

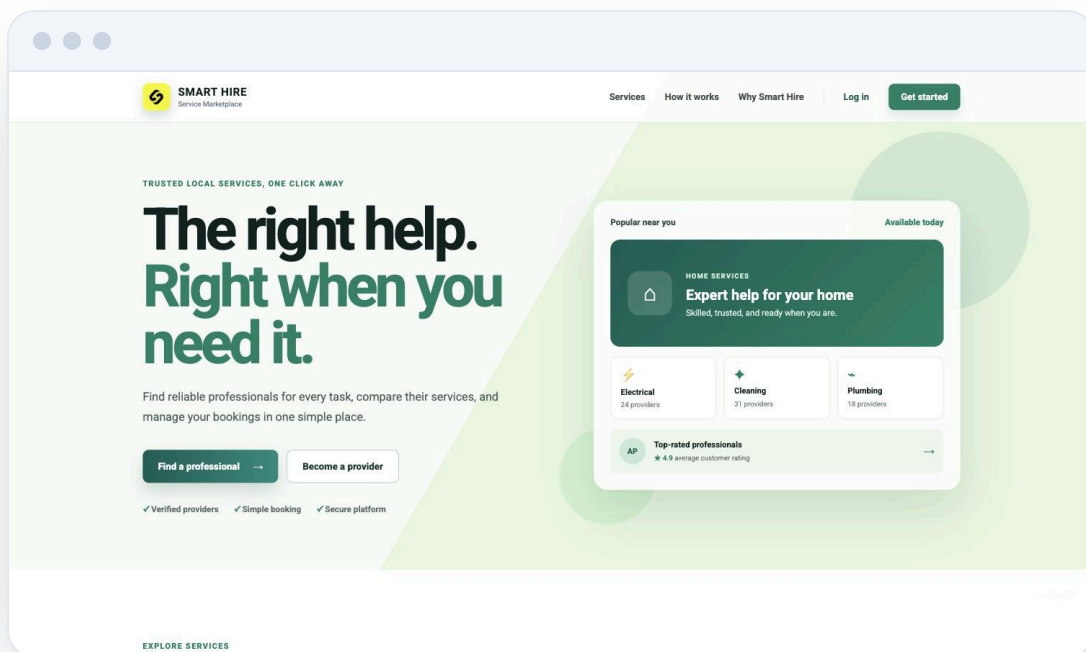
Providers see a workspace built for running a business — not a generic account page.



DISCOVER THE MARKETPLACE

Browsing feels like the landing page promises.

Customers explore providers by category and city; providers define exactly where and what they cover.



Category & city browsing

Services grouped by category (electrical, cleaning, plumbing...) with live provider counts.



Favourites

Customers save services they're considering, without committing to a booking yet.



Provider service areas

Providers set the districts and radius they actually cover.

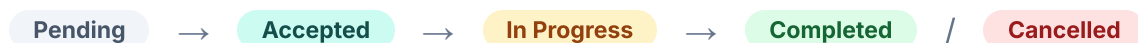


Service listings

Providers manage their own catalogue — pricing, duration, status.

A booking is a real state machine, not a form submission.

Every booking moves through a defined lifecycle, checked against the provider's actual availability, with notifications keeping both sides in the loop.



Weekly availability

Providers set recurring open hours per day; bookings are checked against them.



Notifications

Booking, review and message events surface as in-app notifications for the right person.



Booking-scoped messaging

Customer and provider can message each other within the context of one booking.

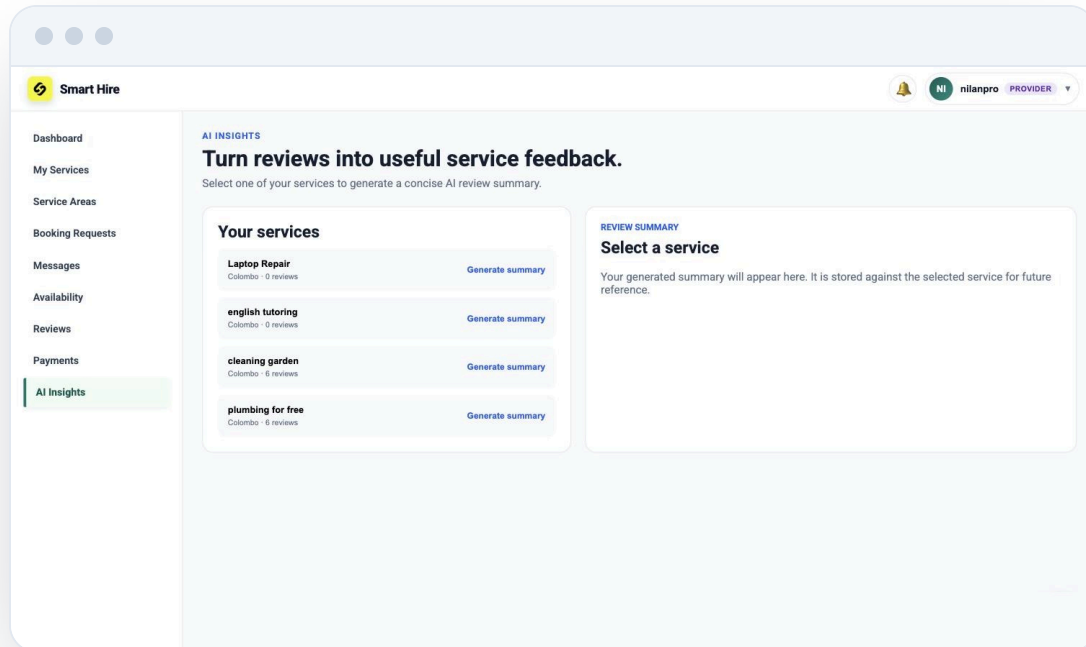


Future-dated bookings

Booking dates are validated server-side, not just in the UI.

AI that reads real reviews, not generic prompts.

An AI service layer turns a service's actual customer reviews into a short, useful summary for its provider, and powers a conversational assistant backed by real conversation history.



Per-service review summaries

Generated on request, stored against the service for future reference.



Persisted conversations

AI chat history is stored per user, not thrown away after the session.



Config discipline

Learned the hard way: an AI feature without a default for its API key can take the whole app down on boot.

Trust and money, both handled as first-class data.

Reviews are locked to one per completed booking at the database level, and every payment carries a real, trackable status.



One review per booking

Enforced by a unique database constraint, not just app logic.



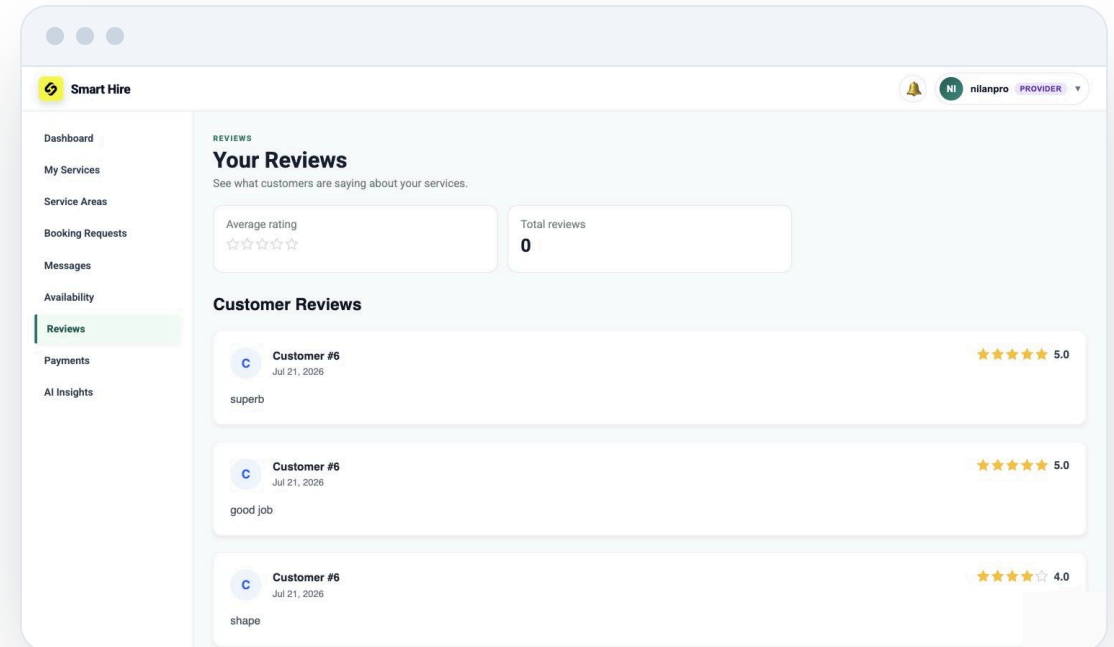
Live rating aggregation

A provider's average rating and review count recalculate on every new review.



Payment lifecycle

Pending → completed / failed / refunded, each with its own transaction record.



Someone has to see the whole marketplace at once.

A dedicated admin role moderates content, manages accounts, and watches the numbers behind the whole platform — separate from the customer and provider experiences.



USERS

Suspend, verify, manage roles



CATEGORIES & SERVICES

Moderate listings platform-wide



REVIEWS

Handle disputed or flagged reviews



PAYMENTS & BOOKINGS

Platform-wide financial oversight

48 automated tests — and 8 real bugs we actually chased down.

Every bug below was reproduced against the running app, root-caused, and fixed — not simulated for this slide.

BUG Admin dashboard revenue/bookings chart was 100% hardcoded fake data

BUG Navbar notification bell always showed a fake "2 unread" for every user

BUG App failed to boot after merge: OPENAI_API_KEY required with no default

BUG 'Verified provider' badge was hardcoded true on every single service card

Tracked in Jira, epic by epic.

The C Clarke Smart Hire Project board (CCSHP) carries the real work breakdown — every epic below maps to a graded deliverable or a working feature area.



BOARD

173

Total work items tracked



PROGRESS

126

Completed this cycle



STRUCTURE

8

Epics across the whole project



BUG TRACKING

8

Real bugs logged & resolved

Four things that changed how we build.

LESSON 19

The zoneless change-detection gap keeps coming back

Plain-field updates inside async callbacks silently never reached the screen — same root cause, three different symptoms, until it became a default habit.

LESSON 11

A `<td>` cannot safely be `display:flex`

One "off-looking" table row led back to a single misplaced flex rule — fixed by moving the flex styling to an inner wrapper.

LESSON 24

Schema-qualify queries against Supabase

An unqualified schema query pulled in Supabase's own internal tables alongside ours, and looked like corruption until it wasn't.

LESSON 21

"Built" isn't the same as "does the one thing that matters"

The AI assistant looked complete but never actually touched the database — a reminder to check the data path, not just the UI.



Thank you.

Questions? — Smart Hire, C Clarke Group